Computer Vision

Perception, Image formation, and
Image identification

Dr Naga Raja G

- **Computer Vision** is a field of artificial intelligence (AI) that enables computers and systems to derive meaningful information from visual inputs like images, videos, and real-time camera feeds.
- It aims to mimic human vision, allowing machines to understand and interpret the visual world.
- Computer vision techniques are widely used in various applications, including object detection, face recognition, image classification, medical image analysis, autonomous vehicles, and augmented reality.

Key Components of Computer Vision:

1. **Perception**:

   This is the initial stage where systems "see" an image or video and begin to understand what is within the scene. It involves detecting and segmenting objects, recognizing edges, shapes, and patterns, and understanding spatial relationships.

2. **Image Formation**:

   Refers to the process of capturing, constructing, and interpreting images. It includes the physics of light, camera optics, and the technology of digital imaging sensors. Understanding image formation is crucial for tasks like correcting lens distortions, calculating depth, and reconstructing 3D models.

3. **Image Identification**:

   Involves recognizing and classifying objects in images, such as identifying faces, animals, or traffic signs. This typically uses machine learning and deep learning models (like CNNs) trained on large datasets to detect specific objects or features.

Applications:

- **Autonomous Vehicles**: Vision systems help recognize pedestrians, obstacles, traffic signals, and other vehicles.
- **Medical Imaging**: Used to detect and diagnose diseases through X-rays, CT, and MRI scans.
- **Facial Recognition**: Identifies individuals by analyzing facial features.
- **Augmented Reality (AR)**: Enhances real-world environments by overlaying digital information.

Perception in Artificial Intelligence

- **Definition:** Perception is the ability to collect, process, and make sense of sensory information, particularly visual data, to recognize objects, environments, and actions.
- **Purpose in AI:** Mimics human vision to allow machines to “see” and interpret surroundings.
- **Components of Perception:**
  - **Image Acquisition:** Obtaining raw data through sensors or cameras.
  - **Preprocessing:** Enhancing image quality, such as noise reduction and normalization.
  - **Feature Extraction:** Identifying fundamental features like edges, shapes, and textures that describe objects within the image.

Key Algorithms in Perception

- **Edge Detection Algorithms:**
  - **Canny Edge Detector:** Uses gradient-based approach to detect edges by smoothing, finding gradients, and tracking along them.
  - **Sobel and Laplacian Operators:** Detect edges by calculating intensity differences between adjacent pixels.
- **Optical Flow:** Measures motion between two frames for applications like object tracking or gesture recognition.
  - **Examples:** Lucas-Kanade, Horn-Schunck methods.
- **Depth Perception Techniques:**
  - **Stereo Vision:** Estimates distance using two camera views.
  - **LiDAR (Light Detection and Ranging):** Measures distances using laser pulses, commonly used in autonomous driving.

Examples of Perception in AI Applications

- **Self-Driving Cars:** Perception systems identify lanes, traffic signs, obstacles, and moving objects, ensuring safe navigation.
- **Robotics:** Perception enables robots to map environments, avoid obstacles, and perform tasks like object manipulation.
- **Surveillance Systems:** Track people and objects, identify abnormal behavior, and enhance security with automatic monitoring.

Image Formation in AI

- **Definition:** Image formation refers to capturing and translating real-world 3D scenes into 2D images using sensors or cameras, taking into account perspective, lighting, and texture.
- **Factors in Image Formation:**
  - **Camera Properties:** Lens focal length, aperture, and field of view determine image sharpness and clarity.
  - **Perspective Projection:** Closer objects appear larger, while distant ones shrink, which must be accounted for to accurately reconstruct scenes.
  - **Lighting and Shading:** Affect color intensity and highlights, crucial for depth perception and recognizing surface properties.

Image Formation Process

- **Projection Models:**
  - **Pinhole Camera Model:** Projects 3D points onto a 2D plane, representing the image as a series of pixel coordinates.
  - **Homography:** Transforms the image to adjust for camera position or angle differences.
- **Color & Texture Representation:** Involves assigning pixel values based on object material, color, and lighting conditions.
- **Lighting Models:**
  - **Lambertian Reflection:** Assumes light is scattered equally in all directions, often used in simple lighting conditions.
  - **Phong Reflection Model:** Accounts for both diffuse and specular reflections to create a realistic appearance of shiny surfaces.

Algorithms in Image Formation

- **Calibration Algorithms:** Adjust camera distortions and calibrate images for accurate shape and size representation.
  - **Example:** Zhang’s calibration, which estimates lens parameters to reduce distortion.
- **Image Transformation Techniques:** Adjust images for consistency or change perspectives.
  - **Homography Transformations:** Useful for aligning or merging images, as in panoramic stitching.
- **Enhancement Techniques:** Improve image quality for better interpretation.
  - **Histogram Equalization:** Enhances contrast by spreading pixel intensity values.
  - **Noise Reduction:** Filters out irrelevant data using techniques like Gaussian or median filtering.

Examples of Image Formation in AI Applications

- **Face Detection Systems:** Normalize lighting and adjust for camera angles, making it easier to identify faces in diverse conditions.
- **Medical Imaging:** Capture clear images for diagnostic purposes, using calibrated settings for consistent quality.
- **Industrial Robotics:** Accurately measure and align parts during manufacturing, relying on precise image formation.

Image Identification in AI

- **Definition:** Image identification involves recognizing objects, people, or scenes within an image by labeling or classifying them.
- **How It Works:**
  - **Feature Detection:** Uses algorithms to learn visual features, such as textures, shapes, or patterns, within an image.
  - **Classification:** Uses pre-trained models to assign labels to objects based on the learned features, enabling recognition of known patterns.

Core Algorithms in Image Identification

- **Convolutional Neural Networks (CNNs):**
  - **Structure:** Multiple layers (convolutions, pooling, fully connected) detect and combine features.
  - **Popular Models:**
    - **AlexNet (**Named after Alex Krizhevsky)**:** Pioneer in CNNs, winning the ImageNet competition in 2012.
    - **VGG(**Visual Geometry Group)**:** Known for its deep, uniform structure, helpful in object recognition.
    - **ResNet(**Residual Network)**:** Uses skip connections, enabling very deep networks without performance loss.
- **Object Detection Networks:**
  - **YOLO (You Only Look Once):** Real-time object detection that processes the entire image in a single pass.
  - **SSD (Single Shot Multibox Detector):** Efficiently detects multiple objects and their locations.
- **Attention Mechanisms:**
  - **Vision Transformers (ViTs):** Use self-attention to focus on important image regions, achieving high accuracy in recognition tasks.

Examples of Image Identification in AI Applications

- **Face Recognition:** Identify individuals in security systems or tag friends in photos on social media.
- **Medical Imaging Diagnosis:** Recognize and classify abnormalities, such as tumors in CT (Computed Tomography) and MRI (Magnetic Resonance Imaging) scans.
- **Autonomous Vehicles:** Detect and classify road signs, vehicles, and pedestrians to navigate traffic.

Bringing It All Together: Integrating Perception, Image Formation, and Identification

- **How the Process Works Together:**
  - **Perception:** Captures basic elements like edges and colors.
  - **Image Formation:** Translates real-world data into processable images.
  - **Image Identification:** Recognizes and labels objects, enabling intelligent decision-making.
- **Example:** Autonomous Vehicle:
  - **Perception:** Recognizes road elements.
  - **Image Formation:** Processes camera data.
  - **Image Identification:** Identifies obstacles, pedestrians, and traffic signals.